

SkillForge AI

10-Day Full-Stack Build Roadmap for a serious portfolio project

Project pitch

SkillForge AI is a full-stack AI learning and interview preparation platform where users upload learning material, generate roadmaps, create flashcards and quizzes, practice mock interviews, chat over uploaded notes using RAG, track progress, and manage learning from a protected dashboard.

Core Stack	What it teaches
Next.js App Router + TypeScript + Tailwind	Frontend architecture, routing, UI systems, responsive design, and deployment-ready code.
Supabase Auth + Postgres + Storage	Authentication, database CRUD, file uploads, user ownership, and backend security rules.
Supabase pgvector + Gemini API	Embeddings, semantic search, RAG, structured AI generation, and fallback logic.
proxy.ts + Upstash Redis	Route protection, role checks, plan checks, redirects, security headers, and rate limiting.
Stripe test mode + Admin + Logs	SaaS logic, usage limits, billing flow, admin control, logging, and debugging discipline.

Golden workflow rule

Do not move to the next phase until the current phase looks good, works correctly, passes build checks, and you can explain what changed.

Required command gate for most phases: `npm run lint`, `npm run typecheck`, `npm run build`. Add `npm run test` from the testing phase onward.

10-Day Roadmap

Day	Focus	Deliverables	Exit Gate
1	Research, PRD, and Rules	RESEARCH.md, PRD.md, PLAN.md, DESIGN_SYSTEM.md, DATABASE_SCHEMA.md, AGENT_RULES.md, SKILLS.md	No code. Clear product, MVP, data model, and agent rules.
2	Scaffold and Design System	Next.js setup, folder structure, theme tokens, base UI: button, input, card, modal, dropdown, tabs, toast, skeleton, empty/error states	Build passes and UI foundation feels consistent.
3	Auth, Profile, and Proxy	Supabase Auth, signup, login, logout, forgot password, profile table, session provider, route protection with proxy.ts	Logged-out users cannot access dashboard. Refresh keeps session.
4	Learning Goals and Roadmaps CRUD	Learning goals, roadmaps, roadmap tasks, progress update, search, filters, ownership rules	User A cannot read or edit User B data.
5	Materials Upload System	Upload PDF/text/image, save metadata, preview, delete, file size/type validation, ownership rules	Bad files blocked. Users only access their own files.
6	AI Generator System	Generate roadmaps, flashcards, quizzes, interview questions, save outputs, fallback mode, usage counter	AI failure does not break the app. Outputs save to DB.
7	RAG Chat Over Uploaded Notes	Extract text, chunk content, store chunks, generate embeddings, vector search, answer using matched context	Answers are grounded in uploaded material.
8	Quiz and Interview Engine	Quiz attempts, scoring, wrong-answer review, weak topic detection, mock interview sessions, progress charts	User can practice, see score, and know weak topics.
9	Admin, Logs, Rate Limits, Plans	Admin dashboard, user list, role/plan changes, API logs, error logs, Upstash rate limit, free/pro limits, Stripe test checkout	Normal users blocked from admin. Rate limit works.
10	Testing, Polish, Deployment	Unit tests, API tests, Playwright auth flow, SEO, accessibility, final README, Vercel deploy	lint + typecheck + test + build pass. App deployed.

Project Feature Coverage

Area	Included features
Frontend	Home, login, signup, dashboard, profile, settings, sidebar, cards, forms, modals, dropdowns, tabs, responsive layouts, loading states, error states, theme, accessibility, SEO basics, performance checks, frontend tests.
Backend	Authentication, database handling, API routes, CRUD, business logic, AI integration, file uploads, exports, email-ready flows, Stripe test billing, admin logic, notifications, validation, error handling, logging, caching, search, security, backend tests.
Middleware / proxy	Protected routes, login checks, redirects, role checks, plan checks, rate limiting, request filtering, API protection, token/cookie checks, security headers, CORS decisions, maintenance mode, logging, bot filtering, access control.
AI/ML	Structured generation, embeddings, vector database, semantic search, RAG chat, weak topic detection, interview question generation, quiz creation, progress recommendations.

Repository branch plan

phase-01-research-prd, phase-02-design-system, phase-03-auth-proxy, phase-04-roadmaps-crud, phase-05-materials-upload, phase-06-ai-generator, phase-07-rag-chat, phase-08-quiz-interview, phase-09-admin-security-billing, phase-10-testing-deploy.

Daily closing checklist

- Does the feature work end-to-end?
- Does it look clean on mobile, tablet, and laptop?
- Did lint, typecheck, and build pass?
- Did I commit on the correct branch?
- Can I explain what changed without reading the code?
- Did I write a short progress note in Notion?

Interview story

In interviews, describe this as a full-stack AI learning platform with auth, protected routes, database CRUD, file uploads, RAG over uploaded notes, quizzes, progress tracking, usage limits, admin panel, logging, tests, and Vercel deployment. That tells recruiters you built a real product system, not only a landing page.